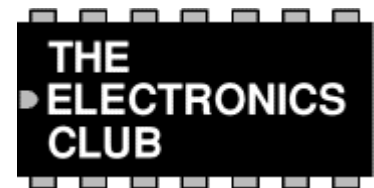


[Home](#) | [Map](#) | [Projects](#) | [Construction](#) | [Soldering](#) | [Study](#) | [Components](#) | [555](#) | [Symbols](#) | [FAQ](#) | [Links](#)

Power Supplies



[Types](#) | [Dual supplies](#) | [Transformer](#) | [Rectifier](#) | [Smoothing](#) | [Regulator](#)

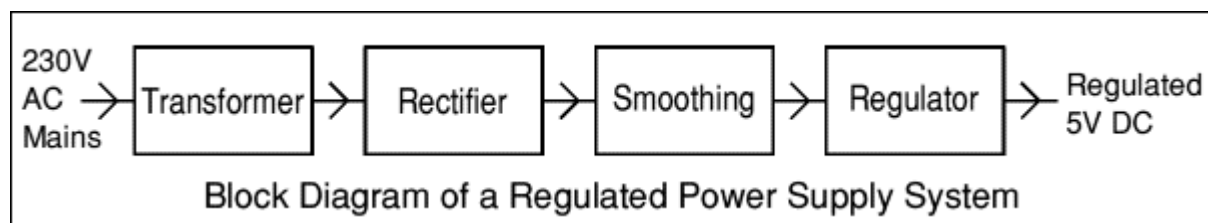
Next Page: [Transducers](#)

Also See: [AC and DC](#) | [Diodes](#) | [Capacitors](#)

Types of Power Supply

There are many types of power supply. Most are designed to convert high voltage AC mains electricity to a suitable low voltage supply for electronics circuits and other devices. A power supply can be broken down into a series of blocks, each of which performs a particular function.

For example a 5V regulated supply:



Each of the blocks is described in more detail below:

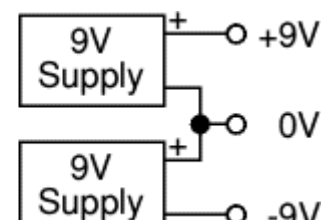
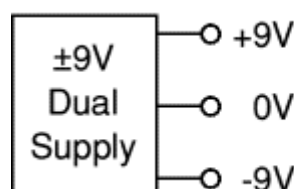
- [Transformer](#) - steps down high voltage AC mains to low voltage AC.
- [Rectifier](#) - converts AC to DC, but the DC output is varying.
- [Smoothing](#) - smooths the DC from varying greatly to a small ripple.
- [Regulator](#) - eliminates ripple by setting DC output to a fixed voltage.

Power supplies made from these blocks are described below with a circuit diagram and a graph of their output:

- [Transformer only](#)
- [Transformer + Rectifier](#)
- [Transformer + Rectifier + Smoothing](#)
- [Transformer + Rectifier + Smoothing + Regulator](#)

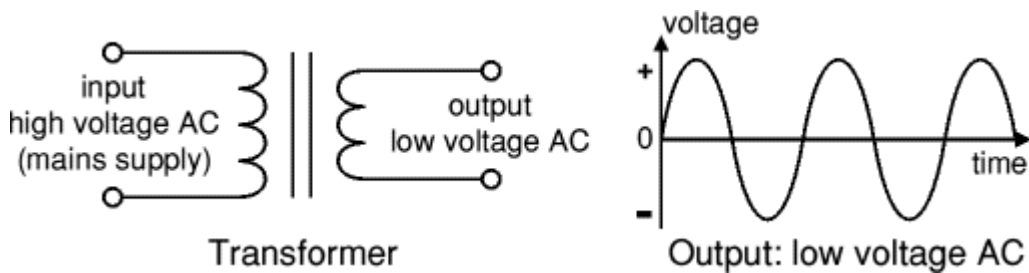
Dual Supplies

Some electronic circuits require a power supply with positive and negative outputs as well as zero volts (0V). This is called a 'dual supply' because it is like two ordinary supplies connected together as shown in the diagram.



Dual supplies have three outputs, for example a $\pm 9\text{V}$ supply has $+9\text{V}$, 0V and -9V outputs.

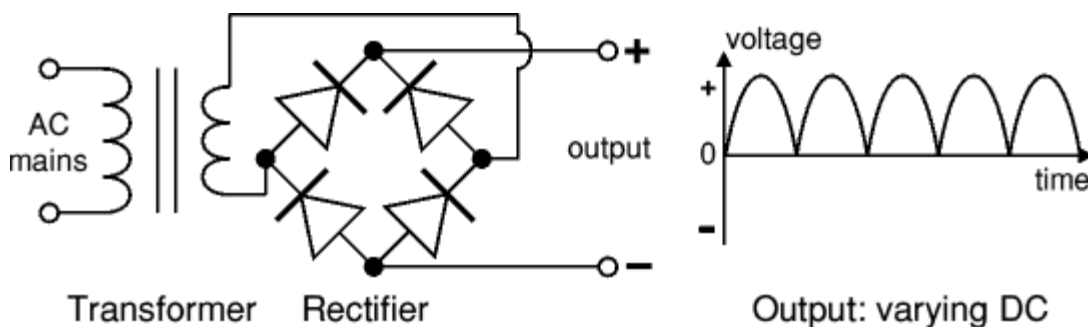
Transformer only



The **low voltage AC** output is suitable for lamps, heaters and special AC motors. It is **not** suitable for electronic circuits unless they include a rectifier and a smoothing capacitor.

Further information: [Transformer](#)

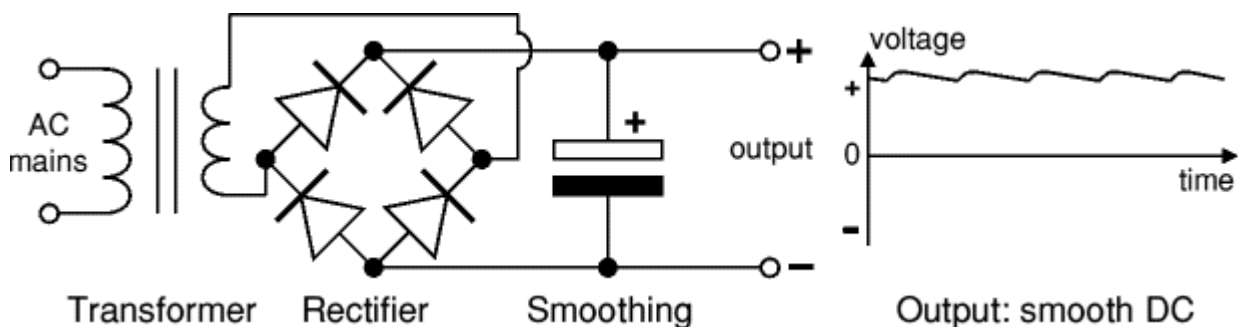
Transformer + Rectifier



The **varying DC** output is suitable for lamps, heaters and standard motors. It is **not** suitable for electronic circuits unless they include a smoothing capacitor.

Further information: [Transformer](#) | [Rectifier](#)

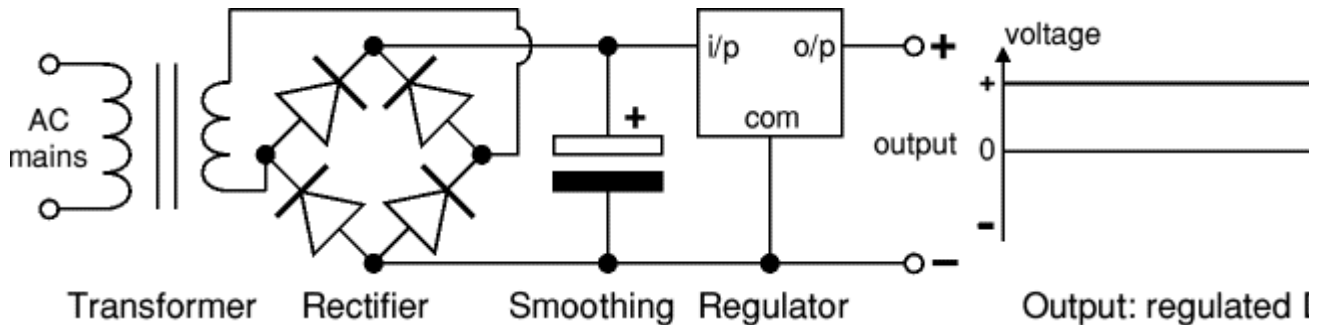
Transformer + Rectifier + Smoothing



The **smooth DC** output has a small ripple. It is suitable for most electronic circuits.

Further information: [Transformer](#) | [Rectifier](#) | [Smoothing](#)

Transformer + Rectifier + Smoothing + Regulator



The **regulated DC** output is very smooth with no ripple. It is suitable for all electronic circuits.

Further information: [Transformer](#) | [Rectifier](#) | [Smoothing](#) | [Regulator](#)

Transformer

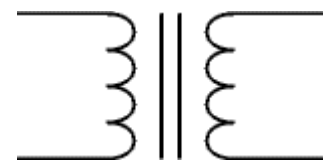
Transformers convert AC electricity from one voltage to another with little loss of power. Transformers work only with AC and this is one of the reasons why mains electricity is AC.

Step-up transformers increase voltage, step-down transformers reduce voltage. Most power supplies use a step-down transformer to reduce the dangerously high mains voltage (230V in UK) to a safer low voltage.

The input coil is called the **primary** and the output coil is called the **secondary**. There is no electrical connection between the two coils, instead they are linked by an alternating magnetic field created in the soft-iron core of the transformer. The two lines in the middle of the circuit symbol represent the core.

Transformers waste very little power so the power out is (almost) equal to the power in. Note that as voltage is stepped down current is stepped up.

The ratio of the number of turns on each coil, called the **turns ratio**, determines the ratio of the voltages. A step-down transformer has a large number of turns on its primary (input) coil which is connected to the high voltage mains supply, and a small number of turns on its secondary (output) coil to give a low output voltage.



Transformer circuit symbol



Transformer
Photograph © [Rapid Electronics](#)

There is more information about transformers on the [Electronics in Meccano](#) website.

$$\text{turns ratio} = \frac{V_p}{V_s} = \frac{N_p}{N_s} \quad \text{and} \quad \text{power out} = \text{power in}$$

$$V_s \times I_s = V_p \times I_p$$

V_p = primary (input) voltage

N_p = number of turns on primary coil

I_p = primary (input) current

V_s = secondary (output) voltage

N_s = number of turns on secondary coil

I_s = secondary (output) current

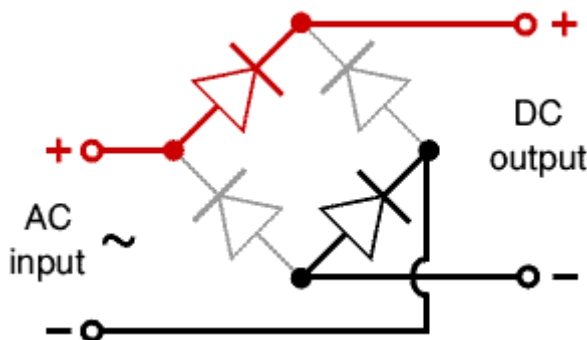
Rectifier

There are several ways of connecting diodes to make a rectifier to convert AC to DC. The [bridge rectifier](#) is the most important and it produces **full-wave** varying DC. A full-wave rectifier can also be made from just two diodes if a centre-tap transformer is used, but this method is rarely used now that diodes are cheaper. A [single diode](#) can be used as a rectifier but it only uses the positive (+) parts of the AC wave to produce **half-wave** varying DC.

There is more information about rectifiers on the [Electronics in Meccano](#) website.

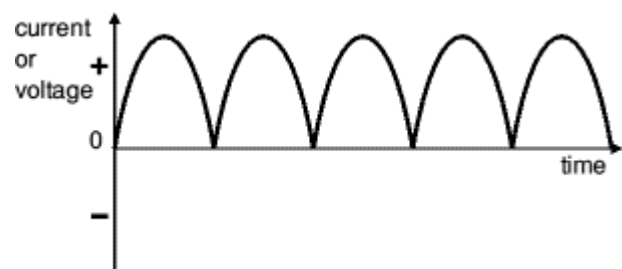
Bridge rectifier

A bridge rectifier can be made using four individual diodes, but it is also available in special packages containing the four diodes required. It is called a full-wave rectifier because it uses all the AC wave (both positive and negative sections). 1.4V is used up in the bridge rectifier because each diode uses 0.7V when conducting and there are always two diodes conducting, as shown in the diagram below. Bridge rectifiers are rated by the maximum current they can pass and the maximum reverse voltage they can withstand (this must be at least three times the supply [RMS](#) voltage so the rectifier can withstand the peak voltages). Please see the [Diodes](#) page for more details, including pictures of bridge rectifiers.



Bridge rectifier

Alternate pairs of diodes conduct, changing over the connections so the alternating directions of AC are converted to the one direction of DC.

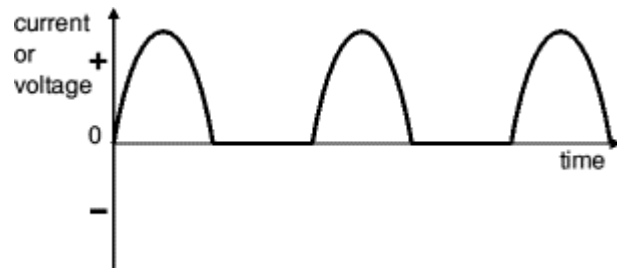


Output: full-wave varying DC
(using all the AC wave)

Single diode rectifier

A single diode can be used as a rectifier but this produces **half-wave** varying DC which has gaps when the AC is negative. It is hard to smooth this sufficiently well to supply electronic circuits unless they require a very small current so the smoothing capacitor does not significantly discharge during the gaps. Please see the [Diodes](#) page for some examples of rectifier diodes.



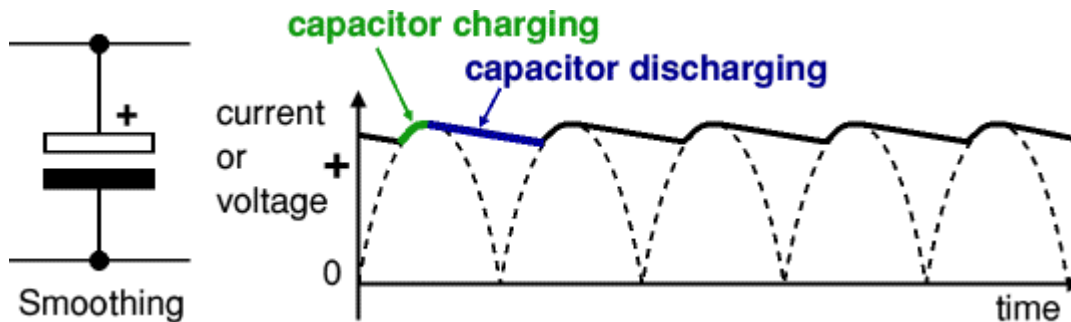


Single diode rectifier

Output: half-wave varying DC
(using only half the AC wave)

Smoothing

Smoothing is performed by a large value **electrolytic capacitor** connected across the DC supply to act as a reservoir, supplying current to the output when the varying DC voltage from the rectifier is falling. The diagram shows the unsmoothed varying DC (dotted line) and the smoothed DC (solid line). The capacitor charges quickly near the peak of the varying DC, and then discharges as it supplies current to the output.



Note that smoothing significantly increases the average DC voltage to almost the peak value ($1.4 \times \text{RMS value}$). For example 6V RMS AC is rectified to full wave DC of about 4.6V RMS (1.4V is lost in the bridge rectifier), with smoothing this increases to almost the peak value giving $1.4 \times 4.6 = 6.4\text{V}$ smooth DC.

Smoothing is not perfect due to the capacitor voltage falling a little as it discharges, giving a small **ripple voltage**. For many circuits a ripple which is 10% of the supply voltage is satisfactory and the equation below gives the required value for the smoothing capacitor. A larger capacitor will give less ripple. The capacitor value must be doubled when smoothing half-wave DC.

$$\text{Smoothing capacitor for 10\% ripple, } C = \frac{5 \times I_o}{V_s \times f}$$

C = smoothing capacitance in farads (F)

I_o = output current from the supply in amps (A)

V_s = supply voltage in volts (V), this is the peak value of the unsmoothed DC

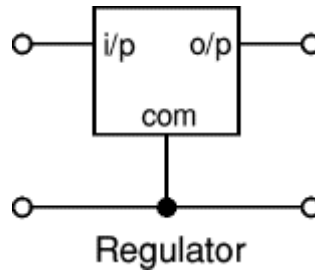
f = frequency of the AC supply in hertz (Hz), 50Hz in the UK

There is more information about smoothing on the [Electronics in Meccano](http://www.kpsec.freeuk.com/powersup.htm) website.

Regulator

Voltage regulator ICs are available with fixed (typically 5, 12 and 15V) or variable output voltages. They are

also rated by the maximum current they can pass. Negative voltage regulators are available, mainly for use in dual supplies. Most regulators include some automatic protection from excessive current ('overload protection') and overheating ('thermal protection').



Voltage regulator
Photograph © [Rapid Electronics](#)

Many of the fixed voltage regulator ICs have 3 leads and look like power transistors, such as the 7805 +5V 1A regulator shown on the right. They include a hole for attaching a [heatsink](#) if necessary.

Please see the [Electronics in Meccano](#) website for more information about voltage regulator ICs.

Zener diode regulator

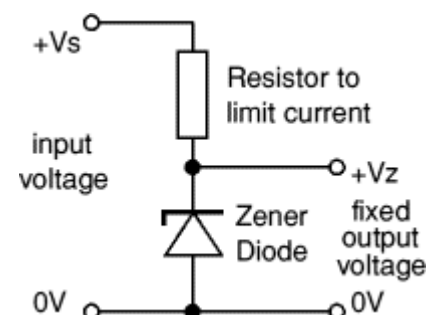
For low current power supplies a simple voltage regulator can be made with a resistor and a zener diode connected **in reverse** as shown in the diagram. Zener diodes are rated by their **breakdown voltage V_z** and **maximum power P_z** (typically 400mW or 1.3W).

The resistor limits the current (like an LED resistor). The current through the resistor is constant, so when there is no output current all the current flows through the zener diode and its power rating P_z must be large enough to withstand this.

Please see the [Diodes](#) page for more information about zener diodes.



zener diode
a = anode, k = cathode



Choosing a zener diode and resistor:

1. The **zener voltage V_z** is the output voltage required
2. The **input voltage V_s** must be a few volts greater than V_z (this is to allow for small fluctuations in V_s due to ripple)
3. The **maximum current I_{max}** is the output current required plus 10%
4. The **zener power P_z** is determined by the maximum current: $P_z > V_z \times I_{max}$
5. The **resistor resistance**: $R = (V_s - V_z) / I_{max}$
6. The **resistor power rating**: $P > (V_s - V_z) \times I_{max}$

Example: output voltage required is 5V, output current required is 60mA.

1. **$V_z = 4.7V$** (nearest value available)
2. **$V_s = 8V$** (it must be a few volts greater than V_z)
3. **$I_{max} = 66mA$** (output current plus 10%)
4. $P_z > 4.7V \times 66mA = 310mW$, choose **$P_z = 400mW$**
5. $R = (8V - 4.7V) / 66mA = 0.05k\Omega = 50\Omega$, choose **$R = 47\Omega$**
6. Resistor power rating $P > (8V - 4.7V) \times 66mA = 218mW$, choose **$P = 0.5W$**

There is more information about regulators on the [Electronics in Meccano](#) website.

Next Page: [Transducers](#) | [Studying Electronics](#)

- [Electronics Club Home Page](#)
- [Site Map](#)
- [Example Projects](#)
- [Construction of Projects](#)
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- [555 Timer](#)
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